MILITARY SPECIFICATION

PROJECTILE, 155MM: XM864
PROPELLANT GRAIN

This Amendment forms a part of Military Specification MIL-P-70505 (AR), dated 28 December 1984, and is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1 Add the following Military Standard:

MIL-STD-453 Inspection, Radiographic.

PAGE 3

3.3 Delete "930 ± 35 cal/g" and substitute "950 ± 50 cal/g".

3.4 Delete in its entirety and substitute the following:

"3.4 Strand burning rate. Strand burning rate tests shall be preformed as shown in TABLE I. Only results from the 70°F, 14.7 psia test shall be used for lot acceptance.

TABLE I. Strand burning rate.

Temperature (± 5°F)	Pressure (psia ± 10%)	Rate (mm/sec)
-50	14.7	INFO
70	14.7	1.00 ± .10
145	14.7	INFO
-50	7.4	INFO
70	7.4	INFO
145	7.4	INFO

3.6.3 Delete in its entirety and substitute the following:

"3.6.3 Internal Defects The propellant grain shall contain no cracks, voids or fissures measuring more than .30 inch in any direction and no foreign material measuring more than 0.20 inch in any direction. The total combined length of all cracks, voids and fissures for the grain shall not exceed .60 inch and the combined length of all foreign material (maximum dimension each particle) shall not exceed 0.40 inch."

AMSC N/A FSC 1320

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

3.7 Delete paragraphs a and b and substitute the following:

"a. Modulus of elasticity of 700 psi minimum per sample average.

b. Ultimate stress of 170 psi minimum per sample average."

PAGE 5

4.3.1 Change strand burning rate specimens from "12" to "18".

PAGE 6

4.3.1 Change "TABLE I" to "TABLE II".

4.3.1 Table II, change strand burning rate samples from "12" to "18". Add the following in the paragraph reference section of the strand burning test.

"(14.7 psia tests only, 6 at each temperature)".

PAGE 10

4.4.2.3 Change title to "Precast Inhibitor, Prior to assembly to grain". Delete special defect A and substitute the following:

"A. Inhibitor prior to bonding to propellant (applies to precast inhibitor only) 100%, 3.6.2, Visual".

PAGE 11

* 4.4.2.4 Change Major Defects 104, 106, and 107 as follows:

Major 104: Change "four" to "two".

Major 106: Change "4" to "2".

Major 107: Change "8" to "4".

Minor 201: Change "location of spacers" to "true position of spacers".

PAGE 12

4.4.2.5 Change Minor 202 from "100%" to ".65%".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

PAGE 13

4.5.1 Delete in its entirety and substitute the following:

"4.5.1 Strand burning characteristics of propellant Along with propellant grains cast from each batch or mix of chemicals, eighteen test strands shall be taken from the same propellant batch and cured in the same curing chamber under the same conditions for the same length of time. The strands shall be tested in accordance with method T803.1 of MIL-STD-286 or an approved alternate (see 6.5). Three strands shall be tested at each of the temperature - pressure combinations in 3.4. If the average of three strands fails to meet the requirements of 3.4, the batch shall be rejected."

PAGE 15

4.5.5 Delete "3.6.2" in the last sentence and substitute "3.6.3". Add the following sentence at the end of the paragraph:

"Equipment and procedures used for radiographic inspection shall be in acordance with MIL-STD-453."

PAGE 16

4.5.10.3 Changed paragraph number to "4.5.6.3".

4.5.10.4 Change paragraph number to "4.5.6.4".

PAGE 17

6.4 Change "SMCAR-LCU-SI" to "SMCAR-FSA-IM".

Custodian:
Army AR

Preparing Activity:
Army AR

(Project 1320-A883)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.